

NAME(S):

### More Lists

Begin the following as paired-programming in class, but be sure to individually finish any problems you don't get to (hand in written answers of **0.25 engagement points**, check website for due date)

### Learning goals / relevant topics

- List methods (Runestone Ch. 9)

### Part 1: list\_micro.py

1. Download the starter file `list_micro.py` from the course website.
2. This problem asks you to write five "micro programs" consisting of just a statement or two.
  - a. Write a statement that assigns a list of 5 integers (any values) to the variable `some_numbers`.
  - b. Write a statement that concatenates the lists `list1` and `list2` and assigns the result to the variable `list1and2`.
  - Note that the starter file assigns some specific values to the variables `list1` and `list2` so that you can test your code. However, the statement you write should not depend on `list1` and `list2` having specific values. Your statement should work for any values that `list1` and `list2` might have. This is also true for all of the following assignments. None of the statements you write should depend on any variables having a specific value.
  - c. Write a statement that computes the sum of the lengths of `list3` and `list4` and assigns the result to the variable `combined_lengths`.
  - d. Write a statement that introduces the variable name `third_element` and assigns the third element of list `list5` as its value.
  - e. Assume that `list6` is a list of numbers. Write a statement that adds the first and last element of `list6` and assigns the result to the variable `first_plus_last`.

### Part 2: minimum.py

1. Download the starter file `minimum.py` from the course website.
2. Write a function called `find_min` that takes a list of numbers and returns the smallest number in the list.
  - Note that the smallest number may not be a unique element in the list; there may be multiple occurrences of that value. If the list is empty, no smallest number can be determined and the function should return `None`. Your definition should not use the built-in `min` function, and the function should not modify the list it is given as a parameter.

3. Now write a function `drop_lowest` that takes a list of scores (numbers) and returns a new list containing all numbers from the first list except for the smallest one. For simplicity purposes if the smallest number occurs more than once, all occurrences should be eliminated.

- Your definition should not use the built-in `min` function. Instead, you should use the `find_min` function you previously defined. This function should not modify the list it is given as a parameter. Add your function to the `minimum.py` file that already contains your definition for `find_min`.

**Prompt: Which function definition will return [12, 15, 18] when called as shown below? Show a trace with memory diagrams using your chosen function definition. Line numbers are provided to help with your trace.**

(lines 1 – 10 hold the `sum` function definition and comments)

```
11 scores = [[1, 2, 3],
12           [4, 5, 6],
13           [7, 8, 9]]
14 sum(scores)
```

a.

```
1 def sum(grid):
2     sums = []
3     for row in grid:
4         sum = 0
5         for num in row:
6             sum += num
7         sums += [sum]
8     return sums
```

b.

```
1 def sum(grid):
2     sums = [0] * len(grid)
3     for row in grid:
4         for i in range(0, len(row)):
5             sums[i] += row[i]
6     return sums
```

c.

```
1 def sum(grid):
2     sum = 0
3     for row in grid:
4         for num in row:
5             sum += num
6     return sum
```